

Scattering part 3

From the previous section we have

$$f(\theta) = -\frac{2m}{\hbar p} \int_0^\infty \sin\left(\frac{pr}{\hbar}\right) V(r) r dr$$

where p is momentum transfer

$$p = \sqrt{4mE(1 - \cos \theta)}$$

Find $f(\theta)$ for the Yukawa potential

$$V(r) = \frac{\beta}{r} \exp(-\epsilon r)$$

By the exponential form of the sine function we have

$$f(\theta) = -\frac{m\beta}{\hbar p} \int_0^\infty \left[i \exp\left(-\frac{ipr}{\hbar} - \epsilon r\right) - i \exp\left(\frac{ipr}{\hbar} - \epsilon r\right) \right] dr$$

Solve the integral.

$$f(\theta) = -\frac{m\beta}{\hbar p} \left[\frac{i}{-ip/\hbar - \epsilon} \exp\left(-\frac{ipr}{\hbar} - \epsilon r\right) - \frac{i}{ip/\hbar - \epsilon} \exp\left(\frac{ipr}{\hbar} - \epsilon r\right) \right]_{r=0}^{r=\infty}$$

Evaluate the limits. The exponentials vanish at the upper limit.

$$f(\theta) = 0 - f(\theta) \Big|_{r=0} = \frac{m\beta}{\hbar p} \left(\frac{i}{-ip/\hbar - \epsilon} - \frac{i}{ip/\hbar - \epsilon} \right)$$

The fractions rationalize as

$$\frac{i}{-ip/\hbar - \epsilon} - \frac{i}{ip/\hbar - \epsilon} = -\frac{2\hbar p}{p^2 + \epsilon^2 \hbar^2} \quad (1)$$

Hence for the Yukawa potential

$$f(\theta) = -\frac{2m\beta}{p^2 + \epsilon^2 \hbar^2} \quad (2)$$

Eigenmath script